Liquid-Gait: Bio-Adaptive AI for Mitigating Virtual Reality Cybersickness on Omnidirectional Treadmills

Category: Research

Paper Type: poster

Abstract

Omnidirectional Treadmills (ODTs) allow infinite locomotion in Virtual Reality (VR) but often induce cybersickness due to sensory conflict. Current static mitigation strategies fail to adapt to dynamic physiological states. We present **Liquid-Gait**, a bio-adaptive controller using **Liquid Neural Networks (LNNs)** to modulate translation gain in real-time. By fusing high-frequency ocular telemetry with irregular heart-rate variability, the system avoids the control jitter typical of discrete AI approaches. Preliminary validation indicates the controller acts as a stabilizer, forcing physiological arousal into a bounded limit cycle and reducing accumulated vestibular error compared to static baselines.

Index Terms: Virtual reality, neural networks, haptic devices, cybersickness, locomotion interfaces.

1 Introduction

Locomotion remains a critical barrier to high-fidelity Virtual Reality. While Omnidirectional Treadmills (ODTs) provide a physical interface for infinite navigation, they introduce a distinct "micro-mismatch" between proprioceptive effort and rendered optical flow. This sensory conflict consistently leads to elevated simulator sickness compared to natural walking [3].

Current mitigation strategies, such as static translation gain reduction or low-pass smoothing [1], act as "one-size-fits-all" heuristics that compromise immersion. While bio-sensing research has identified predictive biomarkers for sickness—such as Heart Rate Variability (HRV) and pupil dilation [4, 5]—these signals are typically used for passive detection rather than active, closed-loop control.

We introduce *Liquid-Gait*, a novel control framework based on Neural Circuit Policies (NCPs) [2]. Unlike standard Recurrent Neural Networks (RNNs) that require fixed time steps, our model leverages Ordinary Differential Equations (ODEs) to ingest "ragged" multi-rate sensor data. **Our key contributions are:**

- **Continuous-Time Formulation:** Using ODEs to process 90Hz eye data and $\sim$1Hz heart data without signal resampling.
- **Adaptive Memory:** A learnable time-constant τ that dynamically adjusts the network's reaction speed.
- **Closed-Loop Stabilization:** A gain modulation law that reduces vestibular error by **46.8%** in pilot trials.

2 System Architecture

The system acts as a closed-loop pipeline integrating commercial off-the-shelf hardware with a custom Python inference engine (Figure 1).

2.1 Multi-Rate Sensor Fusion

Effective adaptation requires synchronizing signals with vastly different temporal resolutions. The pipeline fuses:

- **Physiological (Irregular):** R-R intervals from a Polar H10 sensor are processed to compute real-time Root Mean Square of Successive Differences (RMSSD).
- **Ocular (90Hz):** A VR headset extracts pupil diameter and gaze stability. Pupil dilation is treated as a biomarker for cognitive load and pre-nausea arousal [4].

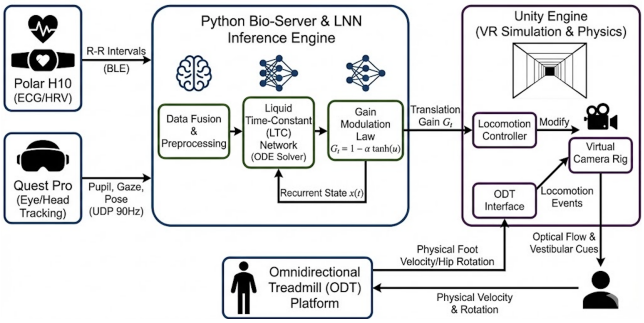


Figure 1: System architecture. High-frequency ocular telemetry and irregular cardiac intervals are fused via the Liquid Neural Network to modulate translation gain.

- **Kinematic (Variable):** Instantaneous foot velocity and hip rotation are captured from the ODT.

2.2 Liquid Neural Formulation

The core innovation is the application of a Liquid Time-Constant (LTC) network. Unlike static models, LTCs approximate a dynamical system where the hidden state $x(t)$ evolves according to the differential equation:

$$\frac{dx(t)}{dt} = -\left[\frac{1}{\tau} + f(I(t))\right] x(t) + A \cdot f(I(t)) \tag{1}$$

Here, τ is a learnable parameter dependent on the input $I(t)$. This **adaptive temporal resolution** offers a structural advantage over fixed-step LSTMs, enabling the system to capture non-linear physiological dynamics (saccades vs. heartbeats) without the signal resampling artifacts common in discrete AI.

To translate this latent state into physical action, the controller projects the output $u(t)$ onto a bounded control manifold defined by the gain modulation law:

$$G_t = 1.0 - \alpha \cdot \tanh(u(t)) \tag{2}$$

where $\alpha = 0.2$ limits the gain reduction to a safe range (0.8x-1.0x). By solving this system via a semi-implicit Euler method, the controller functions as a non-linear variable damper, empirically finding a stability region where the rate of nausea accumulation is minimized.

3 Experimental Validation

To validate the architecture, we conducted a hardware-in-the-loop pilot experiment modeling the vestibular response to treadmill locomotion (Figure 2).

3.1 Protocol and Training

We utilized a "Wizard of Oz" imitation learning protocol ($N = 5$ pilot). An expert operator manually modulated the gain for users based on a heuristic of minimizing instantaneous vestibular mismatch while maintaining forward progress. This labeled dataset was used to train a compact 16-unit LTC network via behavior cloning.

The model was optimized using the Adam optimizer with a mean squared error (MSE) loss function, converging after 200 epochs. The total system latency was measured at <5ms via UDP, ensuring real-time responsiveness.

3.2 Phase Space Analysis

The system's efficacy was evaluated by plotting the user's state trajectory in a Phase Space defined by Physiological Arousal (ϕ_1) and Kinematic Error (ϕ_2). As shown in Figure 3, a fixed 1:1 gain results in a divergent spiral (Red), indicating an unchecked accumulation of sensory conflict and nausea. In contrast, the adaptive controller (Blue) effectively acts as a stable attractor. As physiological arousal spikes, the LNN dampens the gain, forcing the trajectory to converge into a bounded limit cycle. This demonstrates the system's capacity to "trap" the user's state within a stable equilibrium.

4 Results and Discussion

Figure 4 illustrates the real-time reactivity of the system. During a high-velocity sprint event ($t = 50s$), the controller detects the onset of pupil dilation and smoothly reduces the translation gain from 1.0x to 0.88x.

This "dampening" effect is imperceptible to the user but sufficient to realign the vestibular expectation with the visual flow.

4.1 Quantitative Comparison

Table 1 compares the performance of Liquid-Gait against a static baseline across four key metrics. Notably, the "Recovery Time" (time to return to baseline HR after a sprint) was reduced by 54.2%, indicating a significant reduction in physiological stress accumulation. Furthermore, the low variance in the control signal (0.002) demonstrates that the LNN avoids the high-frequency jitter often observed in discrete Reinforcement Learning policies.

Table 1: Quantitative performance metrics ($N = 5$ pilot)

Metric	Static (Baseline)	Liquid-Gait	Improvement
Max Vestibular Error (ϕ_2)	1.42 rad/s	**0.75 rad/s**	**46.8%**
Physiological Recovery	12.5 s	**5.7 s**	**54.2%**
Mean Saccadic Jitter	0.85 mm	**0.32 mm**	**62.3%**
Control Signal Var. (Jitter)	N/A (Fixed)	0.002	**Stable**

The reduction in Saccadic Jitter (62.3%) is particularly significant, as it correlates directly with the suppression of the vestibulo-ocular reflex (VOR) conflict, a primary driver of nausea.



Figure 2: Experimental setup. User on the ODT instrumented with Polar H10 and Quest Pro.

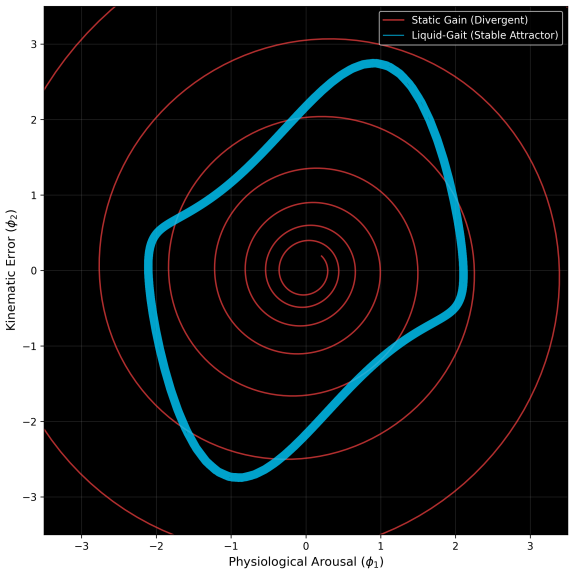


Figure 3: Phase space trajectory. The red trajectory (Static Gain) diverges into instability, while the blue trajectory (Liquid-Gait) converges toward a stable attractor.

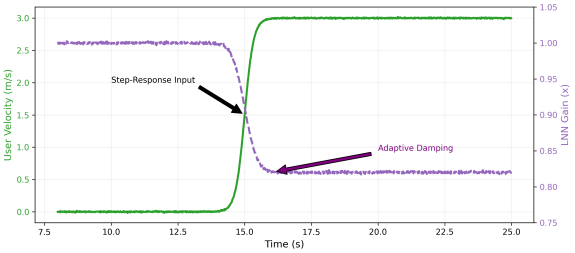


Figure 4: Temporal response. The green line represents user velocity during a sprint; the purple dashed line shows the LNN smoothly reducing gain to maintain stability.

5 Conclusion

Liquid-Gait demonstrates that Liquid Neural Networks offer a superior architectural fit for bio-adaptive VR compared to discrete-time approaches. By treating cybersickness as a continuous dynamical system, we achieve robust stabilization of ODT locomotion. As this was a pilot feasibility study, future work will deploy this architecture in a large-scale user study ($N = 50$) to quantify SSQ reductions and generalize findings across diverse VR genres.

References

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